Three Voices: Uncertainty-Gated Rendering of a Maternal-Health Risk Alarm for Mothers, Community Health Workers, and Clinicians

Maitreya Sapariya*
Department of Information Technology,
A. D. Patel Institute of Technology (ADIT),
Charutar Vidya Mandal (CVM) University,
New Vallabh Vidyanagar, Anand, Gujarat, India

Aditi Patil
Smt. B. K. Shah Medical Institute & Research Centre,
Sumandeep Vidyapeeth (Deemed to be University),
Vadodara, Gujarat, India

Abstract

Three Voices renders one maternal-health risk prediction as three action signals: a decomposition for a clinician, a referral card for a community health worker, and a three-lamp visual with one spoken Hindi sentence for the mother. One band (green, amber or red) is computed from the model's class probabilities. When a high-risk call falls within a fixed top-two margin of 0.15, red is withheld from the health worker and the mother and replaced by amber, which still asks for follow-up; the clinician sees the probabilities unchanged. All text comes from fixed templates; SHAP attributions supply the card's drivers, and no language model runs at inference. On out-of-fold scores from the UCI Maternal Health Risk dataset, the gate derates 11 of 114 high-risk calls; 3 of those 11 are truly high-risk, against 78 of the 103 it keeps. It does not address the 14 high-risk records the model labels low. We also quantify benchmark leakage: 561 of the 1012 usable rows duplicate an earlier row, and keeping them raises macro-recall from 0.580 to 0.859 under the same cross-validation. The evaluation is a formative heuristic inspection by one author; no user study has been run. Five commands regenerate every table and rendering.

Keywords: explainable AI; uncertainty; selective prediction; maternal health; community health workers; risk communication.

1 Introduction

A risk score computed at a rural antenatal visit is read by at least three people. A clinician wants the decomposition and can argue with it. A community health worker wants to know whether to refer this mother and what to write down. The mother wants to know what to do next, and she may not read at all, in which case the score reaches her as a colour and a spoken sentence or it does not reach her.

Most work on explaining a model to several audiences varies how much of the explanation each one gets. That leaves a second question open. When the model is not confident, which assertion should each channel carry?

Consider a case from the dataset we use, scored by a model that did not see it. The model puts the record at 0.233, 0.382 and 0.384 across low, mid and high risk. The top two are

*Corresponding author. Email `maitreyasapariya@gmail.com`. ORCID `0009-0003-9346-3775`.

separated by 0.002, and the model is wrong: the record is true mid-risk. A clinician shown that split can see the call was nearly a tie. The mother, at the end of the same pipeline, has no way to discount anything, and if the rendering follows the argmax alone she is the one who receives an emergency generated by what is, for the model, close to a coin flip.

Our response, which we call Three Voices, makes the abstention decision part of the rendering and leaves the model unchanged. One band is computed per prediction from the class probabilities, and the highest-severity signal is raised only when the model is not on a knife edge. Below the threshold the mother's channel and the health worker's are both derated, while the clinician's is untouched. The derate is red to amber, never red to green. Amber still says *needs follow-up* and still sends her to a health worker. Nothing in this design removes a mother from care, and a reader who takes "not an emergency" to mean "not a concern" has read it backwards.

1.1 What we claim, and what we do not

We contribute three things, and they are different kinds of object. In the terms HCI uses for its own contribution types [35], two are artifact contributions (a rendering mechanism and the pipeline that implements it) and the third is an empirical one, a measurement on the benchmark underneath both.

A mechanism: abstention graded across stakeholder channels. The decision to withhold is taken once, from the model's output, then rendered as a different assertion for each reader rather than as the same assertion softened. The gate is a rule over class probabilities, stated with its grading in Section 4 and reproducible from the committed tables. Out of fold, the gate derates 11 of the 114 high-risk calls, 3 of those 11 truly high-risk, against 78 of the 103 it keeps (Section 4.5).

An artifact: an auditable rendering pipeline. All text comes from fixed templates, with no language model at inference, so every string can be regenerated and checked line by line. We pay for that in fluency and coverage, as Section 2.1 sets out.

A measurement: leakage-aware re-benchmarking of a saturated public dataset. Of the 1012 UCI Maternal Health Risk rows that survive a two-row plausibility filter, 561 are exact duplicates of an earlier row, which can place the same record in a training fold and the test fold of the same split. Scoring the identical model under the identical cross-validation with those rows retained moves accuracy from 0.641 to 0.854 and macro-recall from 0.580 to 0.859. The inflated accuracy lands inside the band the literature reports for this dataset, which is the comparison Section 3.3 sets out in full. Kapoor and Narayanan set out leakage's taxonomy and its place in the reproducibility crisis, and Rosenblatt et al. show the inflation is worst on small datasets, which is our case [13, 29]. We give it a result's billing because a reader comparing our headline number against the published band needs it to make sense of either.

That explanations should be audience-specific was settled before us [2, 12, 14]. Declining to answer under uncertainty has a long lineage in selective prediction and learning to defer, and our fixed-margin rule is a hand-set and cruder instance of it [10, 20]. Fitting a classifier with SHAP attributions to this dataset is saturated [18, 19]. This paper adds nothing to that modelling work, and Section 2.5 sets out why we build on that pipeline as a vehicle. On the audit, our contribution is the measurement. We found no published report of these duplicates (Section 3.3), and we report what they cost without claiming to have found them first.

1.2 What kind of evidence this is

No mother, no community health worker and no clinician has used any of these renderings. Our evaluation is a formative, single-evaluator heuristic critique of our own artifact. It found five

defects (including a driver list that could print a reading arguing against its own prediction under a heading announcing it as a flag), and all five were fixed before this paper was written. It is an instrument that improved the design, not evidence that any tier helps anyone, and Section 5 reports those defects first.

We ask to be judged on the following. The rendering rule and the pipeline that implements it are the object; the artifact is validated by formative inspection against published usability and accessibility criteria; the leakage effect is quantified and reproducible in either direction from a committed table. The data is public, the task is generic maternal-health risk, and nothing here predicts an outcome for a real patient. Whether any of it helps a mother, a health worker or a clinician is a question Section 6.5 takes up rather than one this paper answers.

Section 2 places the work against the two literatures that frame the design and the three that set the standard it has to meet, Section 3 covers the dataset and the leakage measurement, and Section 4 gives the rendering rule, the three tiers and what the gate does across all 451 records. Section 5 reports the evaluation defects first, Section 6 collects what the design cannot yet support, and Section 8 gives the commands that regenerate every artifact here.

2 Related Work

Two literatures frame the design: stakeholder-tailored explanation and selective prediction. Three more set the standard it has to meet: explainable AI for community health workers, risk communication without numeracy, and the published work on this dataset. The five subsections below take them in that order.

2.1 Stakeholder-tailored explanation

Two of our three tiers reproduce a split this literature settled before us. Arya et al. made the point in a title, *One Explanation Does Not Fit All*, and built the toolkit and taxonomy to go with it [2]. Imrie et al. examine the objectives and requirements of different stakeholders in healthcare and argue that they call for different types of interpretability [12]. Kim et al. build and compare stakeholder-tailored explanation interfaces directly, including that same clinician-versus-patient split [14]. Bello et al. propose a *three-level* pipeline from technical explanation to natural language, which is our structural twin [3]. The audience-specific split and tiering as an architecture both predate this work.

Where we diverge from Bello et al. is the mediator. Theirs is a large language model. Ours is a fixed set of templates, selected by the band and filled from the SHAP attributions, so every string a health worker or a mother receives is determined by the case and can be regenerated and audited. Nothing is generated at inference time and there is nothing to hallucinate. We give up fluency and coverage for a rendering an auditor can check line by line.

We name our tiers by role, and Suresh et al. argue that role is the wrong axis: a stakeholder's role determines neither the knowledge they bring nor the interpretability task they face, and they offer a typology that decouples the two [32]. The objection holds. Those names are shorthand, and what varies across the tiers is what each reader lacks, in reading modality, literacy and decision authority; each tier is designed against that deficit. In this deployment role and deficit track each other closely, since the ASHA is a specific, minimally-schooled cadre and the mother may not read at all. Szymanski et al. propose treating role and expertise as separate dimensions, because the field has often conflated them [33]. Where role and the reader's deficit come apart, a deployment would have to select on the deficit.

2.2 Uncertainty communication, abstention, and selective prediction

Our gate declines the highest-severity signal when the top-two class probabilities are separated by less than a fixed margin. That is a hand-set instance of the reject option Hendrickx et

al. survey, and a cruder one: their taxonomy names the case we implement, rejection under *ambiguity*, where two classes are close, as distinct from rejection under novelty [10]. Mozannar and Sontag give the learning-to-defer version, where the model is trained to route a case to a human expert rather than answer it [20]. From Kompa et al. we take the clinical framing that licenses derating a signal instead of issuing it: in medical machine learning, abstaining on high-uncertainty cases is a safety property [16].

What we add sits one step downstream, and Bhatt et al. are the closest prior work to it [5]. They treat uncertainty as itself a form of transparency and consider explicitly how it should be *communicated to different stakeholders*. The difference is narrow: they make the *display* of uncertainty audience-aware, whereas we make the *suppression* of an assertion audience-aware. The mother receives a different assertion, while the clinician receives the split probabilities unmodified. Section 4 gives the rule that grades those assertions. We find no prior work that makes the abstention *rendering* a function of what each stakeholder can interrogate.

2.3 Explainable AI for community health workers in the Global South

Our second tier renders to a community health worker, and this is the literature that sets the standard it has to meet. Okolo et al.'s systematic review of explainable AI in the Global South found 16 papers, of which 3 involved people and 1 attempted deployment with target users [24].

Studies with community health workers in India set our context. Okolo et al. interviewed health workers in rural India about AI-enabled mobile health tools and about explanations specifically [23, 25]. Solano-Kamaiko et al. built and studied an explorable explanation interface with the same cadre [30]. Srinidhi et al. document ASHA Kirana, a deployment in which ASHAs collect data that feeds an algorithm and returns a report for a doctor [31], which is our data flow, minus the tiering and the uncertainty gate.

Two findings in that context bound what our card can be said to do. Okolo et al. found that community health workers struggled to understand the AI explanations they were shown, while still wanting explanations in their tools [25]. Solano-Kamaiko et al. found that explorable explanations, which pair visual heuristics with written text and invite active exploration, helped health workers build a more nuanced understanding of AI and contest its decisions [30]. Comprehension therefore depends on design and has to be tested. Our card has not been tested, which is why a user study is the next step. We propose a rendering rule and evaluate the artifact that implements it.

2.4 Communicating risk without numeracy

Our third tier renders to a reader who may not read, and this literature both warrants that tier and argues against how we built it. Galesic et al. show that icon arrays let low-numeracy readers extract risk information they otherwise miss [8], and García-Retamero and Cokely's systematic review finds visual aids improve risk literacy most among the vulnerable readers our tier targets [9]. Richter et al. add the negative result that matters most for a voice channel: purely verbal communication of benefits and harms performs poorly for patients with limited health literacy [28], which is why our spoken message is paired with a visual rather than shipped alone. Peters et al. collect current best practice for numeric risk [26], and Bradley et al. test risk presentation in the maternity setting [6].

That literature recommends conveying *magnitude* transparently, through an icon array or a visible denominator. Our mother tier strips magnitude entirely and renders an ordinal band plus a single action, which read flatly is what the field advises against. We take the departure on a condition about when a magnitude is the sender's to assert, and state it in Section 4 beside the tier it governs.

2.5 Machine learning on the UCI Maternal Health Risk dataset

We build on the ensemble and SHAP pipeline this dataset has standardised, which puts what is ours downstream of it, in the rendering. Supervised classification with post-hoc attribution here is saturated: Mamun et al. [19], Maisoon et al. [18], Widyawati et al. [34], Rahman et al. [27], Maheswari et al. [17] and Hossen et al. [11] all fit ensemble classifiers to this dataset and explain them post hoc, most with SHAP, and three resample for class imbalance with SMOTE. We use the same pipeline as a vehicle.

Maisoon et al. are our closest methodological neighbour, so we separate the two explicitly [18]. They use the same dataset with a two-row outlier removal, tree ensembles with SMOTE, stratified five-fold cross-validation, SHAP, and the same low-resource framing. Everything upstream of the explanation is substantially their pipeline. What is ours lies downstream: deduplicating before splitting and reporting what that costs (Section 3), gating the rendering on the model's own uncertainty, and rendering the gated output three ways. Widyawati et al. rank blood sugar and systolic blood pressure at the top, as ours do, and Maisoon et al. name blood sugar and blood pressure among the primary determinants [18, 34], which corroborates the attributions rather than adding a finding of ours.

Yesmin et al. are the closest contemporaneous neighbour [36]. They pair ante-hoc fuzzy rules with post-hoc SHAP on this dataset and validate the result with 14 healthcare professionals in Bangladesh, who showed a strong preference for the hybrid explanation (71.4% across three clinical cases). Their question is which explanation format healthcare professionals trust; ours is whether an assertion should be issued at all, and nothing in their pipeline withholds one when the model is uncertain. Their validation also sits on the one tier our gate leaves untouched. Their study does not ask what a health worker or a mother makes of a derated signal.

3 Data and the Model Vehicle

3.1 Dataset

We use the UCI Maternal Health Risk dataset [1] (repository id 863), released under CC BY 4.0. The distributed file contains 1014 records collected in rural Bangladesh, while the repository page lists 1013 instances [1]. Each record has six features (age, systolic and diastolic blood pressure, blood sugar, body temperature and heart rate) and a three-level risk label: low, mid, high. It is public data and no other data is used anywhere in this work. No real patient record, de-identified or otherwise, appears in this paper or in the repository that accompanies it.

The dataset supports *generic maternal-health risk* and nothing narrower. It carries no label for any specific condition, and none of the laboratory, urinary or obstetric-history fields that would let one be inferred. We use the generic term throughout, in the manuscript, in the code and in every rendered artifact, and no tier names a condition.

The label is a risk stratification recorded alongside the readings, not a delivered outcome. Nothing in this paper should be read as a prediction of what happens to any patient, and that constraint binds the third tier hardest: a spoken message naming a diagnosis would assert something the data cannot support. What we demonstrate is an explanation design over a model fit to this public dataset.

3.2 Deduplication, and why it happens before the split

Two preprocessing steps are applied, both before any train/test split, and both chosen so that neither learns a statistic from the data.

First, two records report a heart rate of 7 beats per minute [1]. These are physiologically impossible and are dropped, leaving 1012 rows. Second, the remaining set contains a large number of exactly repeated feature-and-label vectors: 561 of the 1012 rows are exact duplicates

Table 1: The same model and the same cross-validation, scored with exact duplicate rows dropped and retained. Mean ± standard deviation across five stratified folds.

Duplicates	n	Macro-recall	Accuracy	High-risk recall
Dropped	451	0.580 ± 0.027	0.641 ± 0.025	0.723 ± 0.054
Kept	1012	0.859 ± 0.026	0.854 ± 0.026	0.908 ± 0.031

of an earlier row. We drop them, leaving 451 distinct records: 233 low, 106 mid and 112 high, an imbalance of about 2.2:1 between the largest and smallest class.

Both steps are fixed rules rather than fitted transformations, which is why performing them before the split introduces no leakage: nothing about the test fold informs what is dropped. Everything that *does* learn from data, meaning feature scaling and SMOTE where a configuration used it, is fitted inside each cross-validation training fold only, through an `imblearn` pipeline. We considered the alternative, which is to retain the duplicates under group-aware cross-validation and keep more rows. We rejected it on two grounds: it adds machinery a vehicle model does not need, and deduplication makes our number directly comparable to what a reader gets from `drop_duplicates` applied after the heart-rate filter, the order we use.

Deduplication does not resolve every ambiguity in the data. Among the 451 distinct records there remain 35 feature vectors that carry more than one risk label: identical six-value readings recorded against different levels of risk. No model, however specified, can separate rows it cannot distinguish. We retain these rows rather than adjudicating between them, and return to what they imply for the achievable ceiling in Section 6.

3.3 A leakage-aware re-benchmark: what the duplicates were worth

The duplicate rows inflate the scores. A duplicated row can appear in a training fold and in the test fold of the same split, so retaining them lets a model be scored partly on records it has already memorised. We measured that effect directly. Scoring the identical model under the identical five-fold cross-validation twice, changing only whether duplicates were dropped, gives Table 1 (`results/tables/p1_duplicate_leakage.csv`).

Retaining the duplicates moves accuracy from 0.641 to 0.854 and macro-recall from 0.580 to 0.859. The inflated *accuracy*, 0.854, is the like-for-like comparator for the published figures, which are accuracies, and it sits inside the range reported by the papers below; the published range extends higher still. Mamun et al. report 99.51% accuracy on all 1014 records under 10-fold cross-validation [19]. Maisoon et al. remove two implausible outliers ($N = 1012$) and report 83.74% accuracy at a tuned decision threshold of 0.67 [18]. Neither removes duplicate rows [18, 19]. Yesmin et al. report 88.67% on all 1014 records and likewise describe no duplicate handling, though that figure is not a like-for-like comparison: they add a synthetic regional access score to the six clinical features [36].

Leakage is well documented: Kapoor and Narayanan set out its taxonomy and its role in the reproducibility crisis [13], Rosenblatt et al. show that leakage through repeated subjects inflates performance and that the effect is worst on small datasets, which is our situation [29], and Bernett et al. give the practitioner's checklist [4]. Measuring what leakage costs has precedent: Eltawil et al. re-analyse an inflated published result and recover a realistic figure, and Young et al.'s scoping review finds that studies at high risk of leakage report higher accuracy than those with subject-wise splits [7, 37]. We do not claim to be the first to notice these duplicates. We searched for prior documentation of them in this benchmark and found none, but absence of evidence is weak here: a `drop_duplicates` result is the kind of check that is rarely reported. Our contribution here is the measurement: the published band on this dataset is recoverable only when identical rows are allowed to straddle the split, and after deduplication macro-recall

Table 2: Six configurations under the same five-fold stratified cross-validation on the 451 deduplicated records. Mean ± standard deviation across folds. The two random-forest rows tie on the primary metric at 0.580; Section 3.4 gives the tie-break. Mid-risk recall does not exceed 0.302 in any configuration.

		Overall		Per-class recall		
Model	Resample	Macro-recall	Accuracy	Low	Mid	High
Logistic regression	none	0.553 ± 0.056	0.665 ± 0.048	0.948	0.075	0.635
Logistic regression	SMOTE	0.548 ± 0.051	0.587 ± 0.037	0.682	0.283	0.678
Random forest	none	$\mathbf{0.580} \pm 0.027$	0.641 ± 0.025	0.790	0.227	0.723
Random forest	SMOTE	$\mathbf{0.580} \pm 0.040$	0.627 ± 0.039	0.742	0.283	0.713
XGBoost	none	0.558 ± 0.020	0.621 ± 0.017	0.777	0.236	0.660
XGBoost	SMOTE	0.578 ± 0.022	0.627 ± 0.024	0.747	0.302	0.687

on this vehicle is 0.580. It is reproducible in either direction from the committed table. We report deduplicated figures throughout and state the comparison here, up front, so that our lower headline number reads as the result of removing duplicates and not as a weaker model.

3.4 The model is a vehicle

We compared logistic regression, random forest and XGBoost, each with and without in-fold SMOTE, under five-fold stratified cross-validation. Table 2 gives all six configurations, typeset from `results/tables/p1_model_metrics.csv`. The primary metric is macro-recall, with macro-F1, per-class recall, one-versus-rest ROC-AUC and the confusion matrix as support. Macro-recall is the right primary here because the costs are asymmetric: failing to flag a high-risk mother is the error that matters, and headline accuracy on an imbalanced three-class problem conceals that failure.

The final model is a random forest with no resampling, set to `n_estimators=300` and `random_state=42`. It ties the SMOTE-resampled random forest on macro-recall at 0.580. We broke that tie deliberately rather than by argmax: the unresampled model has tighter variance across folds (±0.027 versus ±0.040), the better high-risk recall (0.723), and no dependence on a synthetic-oversampling step. It also lets SHAP run on the six raw clinical features in their own units, which is what makes the ASHA tier's driver list expressible as a template over attributions at all. The selection is a constant in the code, and the script still prints the argmax winner so the tie stays visible.

We take 0.580 as the operating figure for this deduplicated dataset; the rendering question exists because models on data like this are uncertain. Table 2 says where that uncertainty is concentrated. Mid-risk recall is 0.227 ± 0.057 for the selected model and reaches no higher than 0.302 in any of the six configurations, against 0.790 for low risk and 0.723 for high. The mid class is the one the vehicle cannot resolve, and a rendering that reaches a mother must handle the cases it gets wrong.

A forest scored on the records it was trained on sends identical feature vectors to the same leaves, so where those vectors carry two labels, its vote splits between them and the record looks like a near-tie. Out of fold, each such record is usually scored by a fold model that saw its twin with the other label. Of the 71 records on the 35 conflicting vectors (Section 3.2), 68 are scored outside the 0.15 margin and 66 are mispredicted; they produce 10 RED bands, none for a true high-risk record (`results/tables/p1_gate_summary.csv`). For this reason every case in this paper is scored out of fold.

The global attributions come from the final model, fitted to all 451 records: blood sugar is the leading global driver for every class, with systolic blood pressure second, and diastolic pressure and heart rate close to noise (`results/tables/shap_global_mean_abs.csv`). We write "blood sugar and systolic pressure" throughout because the data supports them as drivers;

"the blood-pressure features" would include diastolic pressure.

3.5 Maternal age

The dataset contains very low maternal ages, reflecting adolescent pregnancy in the source population: 95 of the 451 distinct records are under 18 and the minimum is 10, both written to `p1_cohort_summary.csv` by the preparation step. We retain rather than filter these rows, because a deployed system would see these patients, and dropping them would hide how the model treats them. The designated failure case (row 325, Section 4.4) is a 17-year-old whose false high call, out of fold, is driven by her recorded body temperature of 101 °F (SHAP +0.253), with age contributing almost nothing (−0.001). A false high call like this one is why final judgment stays with a human and why the rendering rules of Section 4 exist. All data here is public; no real patient or child record is exposed.

4 Three Tiers from One Prediction

4.1 What the alarm is, and what derating it means

By *alarm* we mean the highest-severity signal a tier is permitted to raise: the red lamp and the urgent spoken message for the mother, the HIGH header on the health worker's card. The rule this section describes withholds that signal when the model is close to a coin flip, and substitutes a middle signal that still says *needs follow-up*.

The direction of that substitution matters most, so we state it first: the derate is red to amber, never red to green. When the model is close to a tie, its output gives no basis for the loudest signal, and the system falls back to the one below it, which still sends the mother to a health worker. Nothing here removes a mother from care.

4.2 The band rule

Each rendering is driven by one band, computed once per prediction from the class probabilities and shared across all three tiers:

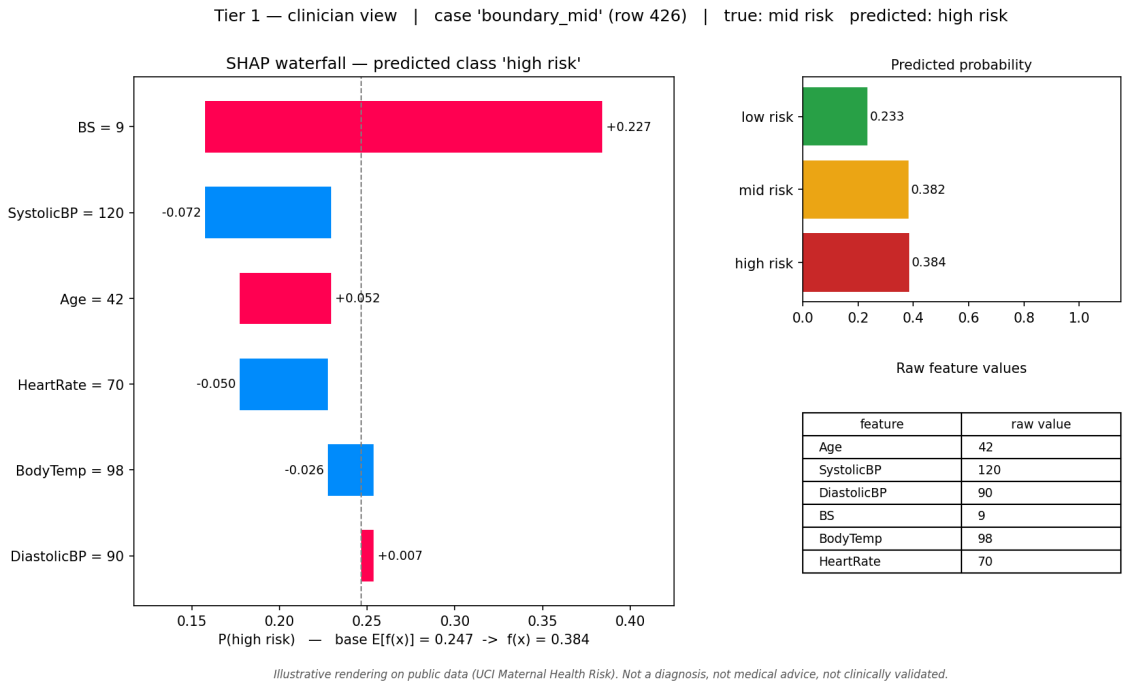
- GREEN if the predicted class is low risk.
- RED if the predicted class is high risk *and* the top-two probability margin is at least 0.15.
- AMBER otherwise, which covers every mid-risk prediction and every high-risk prediction that sits inside the margin.

All probabilities and SHAP values in this section are out of fold: each record is scored by the cross-validation fold model that did not see it (Section 3.4).

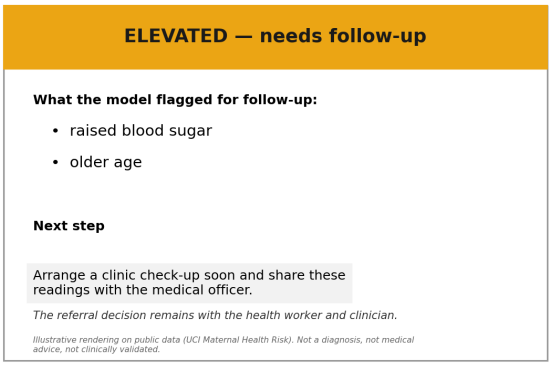
The threshold of 0.15 is a design constant. We did not tune it, and no experiment in this paper supports that value over 0.10 or 0.20; we return to what would be needed to justify it in Section 6.

The gate is one-sided. The margin test guards the high boundary only: GREEN follows the argmax with no margin test at all, so a prediction that is low risk by a hair renders an unqualified all-clear to the mother. That is the opposite of the asymmetry Section 3.4 argues for, where missing a higher-risk mother is the error that matters. Out of fold, 22 low-risk predictions fall inside the margin, and 3 of them are true high-risk records (Table 3). This version does not implement the symmetric extension, derating an ungated near-tie GREEN down to AMBER as the red-to-amber move already does; Section 6.5 takes it up.

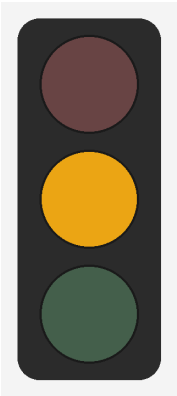
Computing the band once also matters. In the first version of the renderer the health worker's card banded on its own rule and the mother's lamp banded on this one, so a single prediction could receive two different bands. Section 5 describes how that was caught.



(a) Clinician: signed SHAP contributions, all three class probabilities, and the raw readings.



(b) Community health worker: the band header, the two factors pushing toward the prediction, and one next step.



(c) Mother-to-be: amber lit, red dark.

Figure 1: Row 426 of the deduplicated set, scored out of fold and rendered for all three readers. The clinician panel carries the near-tie intact at 0.382 against 0.384. The card and the lamp carry the derated band, and the red they would otherwise have shown is withheld from both. Amber still tells the mother to get checked. All four cases across all three tiers are in the archived repository.

4.3 One prediction, three renderings

Figure 1 shows the same prediction rendered three ways. The case is row 426 of the deduplicated set, a true mid-risk record the model calls high risk. Out of fold, its class probabilities are 0.233/0.382/0.384 across low, mid and high, and the top-two margin is 0.002. Blood sugar (+0.227) and age (+0.052) push toward high risk. Systolic pressure (−0.072), heart rate (−0.050) and body temperature (−0.026) push against it. Diastolic pressure is negligible (+0.007).

Clinician. The full decomposition, unmodified. Signed SHAP contributions per feature with numeric labels, all three class probabilities, and the six raw readings in clinical units. The 0.382 against 0.384 is shown as it stands. A reader who can interrogate the model is given what she needs to do so, including the fact that the call was nearly a tie.

Community health worker. A card carrying the band header, the factors behind *this* prediction, and one next step. Two rules shape it. The factors are local rather than global: naming the dataset's top features on every card would print the same two words for every mother and explain nothing about any of them. And only features that push *toward* the predicted class are listed, ranked by signed contribution, because a card headed "what the model flagged" that lists a reading arguing against its own prediction misleads the person making the referral. On this case a ranking by absolute value would have listed systolic pressure, which argues against the call. The card reads ELEVATED — NEEDS FOLLOW-UP, lists raised blood sugar and older age, and asks the worker to arrange a check-up and share the readings. Every string is a fixed template. The band selects the card header, the next step and the mother's spoken sentence, and the SHAP attributions fill the card's driver list. No language model runs at inference, which is our stated alternative to LLM-mediated tiering [3]: the card can be regenerated and compared line by line, and it contains no generated text.

Mother-to-be. A three-lamp visual and one spoken Hindi sentence. No probability, no percentage, no condition named, and no text required to use it. The tier answers *what do I do next*, not *how likely is this*. Here the amber lamp lights and the message says, in Hindi, that some findings need attention, not to panic, and to get checked soon at her ANM or nearest health centre. The Hindi strings were written and reviewed by the second author, a final-year MBBS student and native Hindi speaker; they are not machine-translated.

Stripping magnitude departs from the risk-communication literature of Section 2.4, which recommends conveying magnitude through an icon array or a visible denominator, and we take that departure on a stated condition. An icon array conveys a magnitude the model can support. At 0.382 against 0.384 there is no such magnitude to convey, and drawing that split as 38 chances in 100 would give the reader least able to discount it a precision the model does not have. When the magnitude is trustworthy it should be shown; when the gate has fired, the model supports only an ordinal band and one action. That is the uncertainty argument of Section 2.2 arriving from the health-communication side, and it remains an argument rather than a result, since whether the band is understood as intended is untested here.

4.4 Derating a false alarm

Row 426 shows the rule behaving correctly on a near-tie between mid and high. Row 325 shows what it is for. That record is true low risk, the model calls it high risk at 0.411/0.148/0.441 across low, mid and high, and the top-two margin is 0.029. It belongs to a 17-year-old with a recorded body temperature of 101 °F, and body temperature drives the call (+0.253). This is the case Section 3.5 designates as the vehicle's failure mode. On the argmax alone she would have received a red lamp and an urgent spoken message. The gate withheld it and both derated channels show amber instead.

Amber on a true-low record is still a false referral. The gate made the error smaller, not correct, and it did nothing about the underlying misprediction, which the clinician tier carries unmodified here as it does on every case. It changed only what the mother was told.

4.5 The gate across all 451 records

Table 3 gives the counts. The gate derates 11 of the 114 high-risk calls. Of the 103 calls it keeps as red, 78 are true high-risk records; of the 11 it derates, only 3 are. Eleven calls are too few to

Table 3: What the gate does to the 451 records, scored out of fold at a margin of 0.15. Upper block: the 114 high-risk predictions, by true label and by the band they receive, and the 22 low-risk predictions inside the margin, which the one-sided gate does not test. Lower block: the 112 true high-risk records, by the band they receive.

	True low	True mid	True high	Total
Predicted high	11	22	81	114
Kept RED	8	17	78	103
Derated to AMBER	3	5	3	11
Predicted low, inside margin (no gate)	14	5	3	22
	Red	Amber	Green	Total
True high-risk records	78	20	14	112

estimate that difference precisely, and this is one dataset. The 3 derated high-risk records are the cost of the rule: they receive amber instead of red, and amber still routes them to a health worker.

The number of red bands changes little with the threshold: 105, 103 and 102 at margins of 0.10, 0.15 and 0.20.

The gate leaves 14 true high-risk records with a green band. Only 3 of them sit inside the margin, among the 22 low-risk near-ties of Section 4.2. The other 11 are outside the 0.15 margin, with margins from 0.230 to 0.880. Overall, 98 of the 112 true high-risk records receive a band that routes them to follow-up.

5 Formative Heuristic Evaluation

We rated the rendered artifacts against WCAG 2.1 and Nielsen's usability heuristics: 16 criteria across three tiers, 48 cells, each grounded in a named detail of a file on disk. Contrast ratios are computed at run time from the palette constants rather than typed in by hand.

Both instruments are inspection methods, each with a stated procedure and a literature on what it does and does not catch. Heuristic evaluation was introduced as a way for evaluators rather than users to find usability problems in an interface design [22], and we rate against Nielsen's revised heuristics [21]; conformance inspection against the WCAG 2.1 success criteria is the corresponding instrument on the accessibility side [15]. Neither substitutes for users.

This is a single-evaluator, author-conducted, formative assessment. It is not user testing and it is not validation. Nothing in it is evidence that the tiers help anyone. What it did was find defects in our own artifact, which is what we report.

5.1 What it found

The first pass ran against the original renderer, on cases selected from in-sample probabilities. It returned four Fail ratings, three of them on the health worker's card, arising from five distinct defects. All five were fixed and the tiers re-rendered, with the cases now scored out of fold (Section 3.4). The ratings below and in Table 4 describe the corrected artifacts.

Sign-blind driver list. Features were ranked by absolute SHAP magnitude with no sign filter, so a reading that argued *against* the prediction could be printed under a heading announcing it as a flag. On the original false-positive case the card listed low blood sugar, whose contribution to the predicted class was -0.085. The routine card was worse: it listed the two features that were the model's reasons for saying *low* under that same heading, above a next step telling the

Table 4: All 48 cells: 16 criteria rated against each tier, on the corrected artifacts. • pass, ○ partial, × fail, – deferred. The per-cell rationale for every rating is in `results/tables/heuristic_matrix.csv`.

Criterion	Clinician	ASHA	Mother
WCAG 2.1			
1.1.1 Text alternatives for non-text content	○	•	○
1.4.1 Colour not the only cue	•	•	○
1.4.3 Contrast (minimum)	•	•	•
1.4.5 Images of text	×	○	•
2.1 / 2.4 Keyboard, focus order, navigation	–	–	–
3.1.1 / 3.1.2 Language of page and of parts	•	○	•
3.1.5 Reading level / plain language	○	○	•
3.2.4 Consistent identification	•	•	•
4.1.2 Name, role, value	–	–	–
Nielsen's heuristics			
1. Visibility of system status	•	•	•
2. Match between system and the real world	•	•	•
4. Consistency and standards	•	•	•
5. Error prevention	•	•	•
6. Recognition rather than recall	○	•	○
8. Aesthetic and minimalist design	○	•	•
10. Help and documentation	○	•	○

reader to arrange a check-up. This is the defect that could have put a wrong cue in front of the person making a referral, and no model metric could have shown it, because metrics never read the card.

Two bands for one prediction. The health worker's card and the mother's lamp banded independently, so the confident high-risk case rendered an amber card beside a red lamp: one prediction with two identities. The band is now computed once and passed to both.

Unmeasured header contrast. White header text was hardcoded; on the amber header, which is the state for two of the four rendered cases, it measured 2.12:1. Ink is now chosen per band by measured contrast, worst case 5.14:1, and a self-check asserts it.

Direction words without a deadband. A blood sugar of 7.7 against a median of 7.5 was worded as "raised". The word now attaches only outside 0.25 IQR of the median.

No provenance on the mother's transcript. The transcript carried none, while both other tiers printed the disclaimer. It now records case, band and disclaimer under an explicit written-not-spoken rule; the spoken line is unchanged, because appending provenance to every message would bury the one action the tier exists to deliver.

5.2 What it could not settle

Table 4 gives the full matrix. Across the 48 cells the distribution after those fixes is 29 pass, 12 partial, 1 fail and 6 deferred. The count is a description of coverage and not a score, and the five defects above carry more weight. The single failure is the clinician tier shipping as an image of text. The six deferred cells are all keyboard operability and assistive-technology semantics, which a static PNG cannot exercise and a deployed application would.

Read by column, each tier has criteria rated partial (clinician 5, ASHA 3, mother 4). Two of those partial ratings sit where a single universal rendering would have to compromise: the

clinician view needs chart and English literacy, the mother's view needs hearing, and neither gap closes inside its own tier without destroying what that tier is for. Read by row, 14 of the 16 criteria are rated pass in at least one tier and 6 in all three; the remaining 2 of the 16 (rows 2.1/2.4 and 4.1.2) are deferred in every tier.

None of this tells us whether a health worker or a mother understands what she is handed. Studies with community health workers in India show that how well they understand an AI explanation depends on its design [25, 30], so the answer cannot be assumed. Studies with mothers, health workers and clinicians are the next step (Section 6.5).

6 Limitations and Future Work

6.1 Evaluation without users

No mother, health worker or clinician has seen any of these renderings. That is the limitation the rest depend on, and it bounds what the paper is allowed to conclude: we have a design and a formative critique of our own artifact, not evidence that the design works.

Four claims sit downstream of it. Whether a health worker reads the card as intended is assumed and not shown; the nearest studies find that health workers can struggle with AI explanations and that the design of an explanation changes how well they understand it [25, 30]. Whether withholding magnitude from the mother helps or harms her is an argument from the model's uncertainty, and the icon-array paragraph of Section 4.3 makes it, but the risk-communication literature it departs from is empirical and our departure is not. The 0.15 margin is a design constant with no experiment behind it, as Section 4 states. And role remains a proxy: we design against what each reader lacks in modality, literacy and decision authority, but recruit nobody to check the proxy holds, which is the substance of Suresh et al.'s objection to role-based tiering [32].

All four resolve the same way, and Section 6.5 says how. Everything here was produced and inspected on a workstation: no field claim, no claim about performance under real clinic conditions, and no claim that these renderings have reached anyone by any delivery channel.

6.2 What one public dataset can support

The work uses the UCI Maternal Health Risk dataset and nothing else. It is public, carries generic three-level risk labels, and names no clinical condition, so neither do we. Nothing here is clinically validated and nothing predicts an outcome for a real patient.

Two properties bound the vehicle. Macro-recall is 0.580 after deduplication, the leak-free figure (Section 3.3); and 35 feature vectors, covering 71 of the 451 records, carry more than one risk label, an ambiguity no model can resolve from six features. That is a ceiling on the data, not a shortfall in the classifier.

One assumption inside the gate deserves naming. The margin rule thresholds a difference between random-forest class probabilities, which are vote fractions and are not calibrated. We treat 0.15 as a design constant on an ordering, not a statement about probability.

The gate also has only one side. It tests the margin at the high boundary and not the low one (Section 4). Out of fold, 22 low-risk predictions sit inside the margin and render an unqualified GREEN to the mother; 3 of them are true high-risk records (Table 3).

The most serious error mode is a missed alarm. Out of fold, 14 true high-risk records receive GREEN: 3 inside the 0.15 margin and 11 outside it, with margins from 0.230 to 0.880. Neither the current gate nor its symmetric extension would reach them.

6.3 Who the renderings still exclude

Each tier assumes something about its reader.

The mother's visual carries no rendered text, so a deaf or hard-of-hearing mother gets nothing from it; the spoken sentence is her only channel and she cannot hear it. A rendered caption or an SMS companion would close this, but it reopens the literacy assumption the tier exists to avoid, so it needs user input rather than a unilateral design call. Colour is the lamp's primary cue. Position, the lit-versus-unlit luminance step and the spoken line are real redundancies, but a reader who is both colour-blind and deaf is left with position alone, which assumes the traffic-light convention. The clinician view needs chart literacy and English. The health worker's card is English-only; every string is a fixed template constant, so localising it is substitution deferred to deployment. All three tiers ship as rasterised images of text.

6.4 The voice is a stand-in

Audio is synthesised with a general-purpose engine reached over the network. It is not tuned for Indian-language health speech, handles acronyms unpredictably, and needs connectivity the setting this design describes may not have. Deployment-grade Indian-language synthesis (AI4Bharat, Bhashini) is the stated path. What we treat as the artifact is the Hindi source string, not the audio file.

6.5 Future work

Studies with mothers, community health workers and clinicians are the first item and the one the others wait on; they are gated on ethics clearance, which this preprint does not have and does not claim. Two changes to the rule are specified enough to name now. The gate should extend to the low boundary, derating an ungated near-tie GREEN to AMBER as the red-to-amber move already does. On these data it would move the 22 near-ties of Section 6.2 to AMBER; we report that count, and the extension is not implemented. The forest's vote fractions should also be calibrated, after which the threshold is worth re-asking. A third direction is an explorable version of the health worker's card, which the findings of Solano-Kamaiko et al. [30] suggest. Sections 6.3 and 6.4 name their own paths.

7 Conclusion

We built a rendering pipeline that takes one maternal-health risk prediction and issues three different assertions from it, one per reader, withholding the highest-severity signal from the two channels that cannot interrogate the model whenever the top-two class probabilities fall inside a fixed margin. All text comes from fixed templates, with no language model at inference, so every string can be regenerated and checked line by line. We also re-benchmarked the dataset underneath it: 561 of the 1012 usable rows are exact duplicates, retaining them across the split moves macro-recall from 0.580 to 0.859, and 0.580 is the figure this vehicle earns once they are dropped.

Row 325 is the case that shows what the rule is for. Out of fold, the model calls this true-low record high risk on a top-two margin of 0.029, and the mother receives amber where the argmax alone would have sent red. That is still a false referral. The gate made the error smaller, not correct. Across all 451 records, the gate derates 11 of the 114 high-risk calls. Three of those 11 are truly high-risk, against 78 of the 103 calls it keeps as RED. Whether any of this helps a mother, a health worker or a clinician is untested here and gated on the ethics clearance Section 6 describes.

8 Reproducibility

Every figure, table and rendered artifact in this paper regenerates from five commands.

`python -m src.data` Fetches the dataset once from the UCI repository, caches it to `data/raw/`, and reports row count, duplicate count and missing values without modifying anything. Every later run reads the cache, so the rest of the pipeline needs no network.

`python -m src.model` Runs five-fold stratified cross-validation over three classifiers with and without SMOTE, fitting scaling and resampling inside each training fold. Writes five CSVs: metrics, leakage and cohort tables (`p1_model_metrics.csv`, `p1_duplicate_leakage.csv`, `p1_cohort_summary.csv`, which carries the deduplication and cohort counts), the out-of-fold score of every record (`p1_oof_predictions.csv`) and the gate counts (`p1_gate_summary.csv`). Saves the selected model.

`python -m src.explain` Computes global mean absolute SHAP per feature per class from the final model fitted to all 451 records, and local SHAP for the four criterion-selected cases from the fold model that held each one out. Writes three CSVs and five figures.

`python -m src.render` Renders the three tiers. It reads the saved SHAP tables rather than the model, so the renderer and the explainer cannot drift apart. `--case all` does all four cases.

`python -m src.evaluate` Rates the rendered artifacts across the 48 cells and writes `heuristic_matrix.csv` and `heuristic_evaluation.md`. Contrast ratios are computed from the renderer's own palette constants at run time.

8.1 Determinism

One seed, fixed at 42 in `src/config.py`, is applied to Python's hash randomisation, the standard library generator and NumPy by a `set_seeds()` call at the top of each script that touches randomness. The loader contains no random component and needs none. The out-of-fold scoring that produces every case probability and gate count runs single-threaded, so the forest's floating-point sums are added in a fixed order. All eleven packages in `requirements.txt` are pinned to exact versions, and `requirements-lock.txt` records the full environment of 35 packages. The pins require Python 3.12, since the NumPy and scikit-learn releases named in `requirements.txt` predate the 3.13 and 3.14 wheels and will not build on a newer interpreter. The dataset is cached after first download rather than re-fetched. A clean run from a fresh clone (Windows 11, Python 3.12, the locked environment, the dataset downloaded afresh) reproduced all 17 PNGs byte for byte and all 18 CSV, Markdown and text outputs with identical content; the text outputs differ from the committed copies only in line endings, which follow the platform. The freshly downloaded dataset was byte-identical to our cached copy.

8.2 The exception

Speech synthesis is not ours to seed. The Hindi audio is produced by a remote service, and re-running the renderer on unchanged input returns files that are sometimes byte-identical and sometimes not. The determinism guarantee above covers our pipeline and stops at that endpoint.

What is deterministic, and what we therefore treat as the artifact of record, is the Hindi source string. It lives in `results/tables/tier_mother_<case>_hi.txt`, it is committed, and it reproduces exactly. The MP3 is a rendering of that string.

One operational consequence is worth recording, because we were caught by it. Re-rendering over an existing audio file can leave the previous file in place while still reporting success and updating its timestamp. A cold run must empty `results/audio/` before re-rendering, or it can certify audio it did not regenerate.

8.3 Artifacts

The dataset is public and CC BY 4.0. Code, all rendered artifacts for four cases across three tiers, the cross-validation and SHAP tables, and the evaluation matrix are archived at `https://doi.org/10.5281/zenodo.22252076`, which is the concept DOI and resolves to the latest release; development history is at `https://github.com/stiFFLer-codes/three-voices`. The paper cites the committed tables directly, so any number in it can be checked against the file that produced it. The leakage comparison with and without duplicates (Table 1) is one of those files rather than a separate analysis: `p1_duplicate_leakage.csv` is written by the same command that writes the main metrics table, from the same cross-validation, so the comparison re-runs whenever the model does.

Declarations

Funding

No funding was received for conducting this study.

Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Ethics Approval

This study involved no human participants and no animals. It is a secondary analysis of the UCI Maternal Health Risk dataset, which is public, released under CC BY 4.0, and contains no identifiable personal data. Ethics approval was therefore not required and none was sought. The studies with mothers, community health workers and clinicians that this paper names as its next step are gated on ethics clearance that has not been obtained and is not claimed anywhere in this work.

Consent to Participate and Consent to Publish

Not applicable. No human participants were recruited and no individual person's data is reported.

Data and Code Availability

We use the UCI Maternal Health Risk dataset [1] (repository id 863), public and released under CC BY 4.0, at `https://doi.org/10.24432/C5DP5D`. No other data is used and no real patient record appears in this work.

All code, the rendered artifacts for all four cases across all three tiers, the cross-validation and SHAP tables, and the evaluation matrix are archived at `https://doi.org/10.5281/zenodo.22252076`. That is the concept DOI and resolves to the latest release; development history is at `https://github.com/stiFFLer-codes/three-voices`. Every figure and table in this paper regenerates from the five commands listed in Section 8.

Author Contributions

Maitreya Sapariya Conceptualization, Methodology, Software, Formal analysis, Writing – original draft.

Aditi Patil Hindi authoring and review, clinical domain input, ASHA-tier design input, Writing – review & editing.

Use of Generative AI in the Preparation of this Manuscript

The authors used an AI assistant (Claude, Anthropic) to help write and revise the code and to draft and revise the text of this manuscript, working under author direction. The research design, the analyses, every reported number and the selection and verification of every citation were determined and checked by the authors. The authors approved the final wording and take full responsibility for the content. This is a statement about how the paper was written, not about the system it describes. In the rendering pipeline of Section 4, all text comes from fixed templates, with no language model at inference.

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